

Pretham Voruganti

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EDUCATION

University of Maryland, College Park

M.S. Applied Machine Learning

College Park, MD

Aug. 2026 – May 2028 (Expected)

Purdue University

B.S. Cybersecurity, GPA: 3.69/4.00

West Lafayette, IN

Aug. 2023 – May 2026

Selected coursework: Applied Machine Learning, Database Fundamentals, Advanced Programming, Statistics, Cloud Engineering, Network & System Administration

TECHNICAL KNOWLEDGE

Programming & Backend: Python, TypeScript, JavaScript, Java, C++, C#, SQL, Bash; Node.js/Express, REST APIs, PostgreSQL, Redis/BullMQ, asynchronous queues, RBAC

Machine Learning & Data: Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, statsmodels, Matplotlib, Power BI; anomaly detection, classification, statistical testing, Monte Carlo simulation, model validation

Cloud & Distributed Systems: AWS (EC2, RDS, S3, Lambda, IAM), Docker, Kubernetes, Terraform, GitHub Actions, Linux; concurrency, multithreading, fault recovery, CI/CD, data structures and algorithms

Security & Observability: Prometheus, network/packet analysis, log and telemetry analysis, vulnerability assessment, TCP/IP, raw sockets, MQTT, Socket.IO; CompTIA Security+, AWS Cloud Practitioner, AWS AI Practitioner

WORK EXPERIENCE

Engineer Intern | Mastercard

May 2025 – Aug. 2026

- Built Python automation for high-volume security telemetry and fraud-signal analysis, transforming noisy event data into repeatable workflows for correlation, triage, validation, investigation, and technical reporting.
- Evaluated ML-based threat/fraud classification approaches, validated analytical outputs, and communicated findings to technical and leadership stakeholders in a regulated financial environment.
- Improved repeatability of recurring investigations by structuring analytical outputs and reducing manual analysis steps; earned multiple awards for innovation and execution.

IT Security Analyst Intern | NSG Info, Alexandria, VA

May 2024 – Sep. 2024

- Automated Python log monitoring and correlation across infrastructure events, identifying abnormal behavior while supporting vulnerability assessment, remediation, and Linux/network troubleshooting.
- Summarized infrastructure and security trends through Power BI and technical reporting, helping translate event data and assessment findings into clear remediation follow-up.

PROJECTS

Smart Food Delivery Locker System - Full-Stack / Distributed Systems

2024 – Present

- Built TypeScript/Node.js services around PostgreSQL, Redis/BullMQ, REST APIs, AWS, authentication/RBAC, Stripe payment flows, and Bluetooth-enabled lockers to coordinate asynchronous delivery workflows.
- Designed concurrency controls, idempotent retries, state recovery, and failure isolation for background jobs and device-facing operations; instrumented system behavior with Prometheus metrics and structured operational data.
- Developed Python analytics to forecast locker demand and examine routing/utilization patterns, using application and delivery data to improve operational decisions and troubleshoot system behavior.

Network Intrusion Detection System - Python / Statistical Detection

2026

- Engineered a 1,900-line Python/Scapy packet-processing system using raw sockets and multithreading, sustaining 1,000+ packets/second at under 1 ms analysis latency with structured JSON/CSV telemetry.
- Implemented 12 signature rules plus 3-sigma/30% anomaly thresholds and severity triage, separating capture, detection, and telemetry paths for repeatable testing, debugging, and investigation.

AI-Powered Lease Analysis Platform - Applied ML / Decision Pipeline

2025 – Present

- Shipped a five-stage normalization, pattern, anomaly, state-law, and conditional-model pipeline across HUD data from 51 jurisdictions, reducing per-analysis cost from \$0.82 to \$0.15 and improving margin from 67% to 92%.
- Routed routine cases through deterministic, testable stages before model inference for ambiguous inputs, improving data quality, model-use efficiency, and failure analysis across the decision pipeline.

RESEARCH

Multiscale Spatial Statistics & Computational Modeling

2025 – Present

- Developed covariance-aware chi-square methods and logarithmic null models; ran 1M+ Monte Carlo simulations across 8 validation experiments measuring false-positive calibration, statistical power, sensitivity, and robustness.
- Benchmarked against DBSCAN and Ripley's K, observing higher detection power in sparse-cluster settings; extended validation to large scientific datasets including MICrONS (8.2M synapses) and MERFISH (1.37M cells).
- Authored five technical manuscripts documenting methodology, experimental design, validation results, assumptions, and limitations across the research program.